

Lecture 26: Oct 22

Last time

- Moment Generating Function

Today

- Common Discrete Distributions

Common Discrete Distribution

Why parametric models?

- *Parametric models* or *distribution families* have a specific form but can change according to a fixed number of parameters.
- The objective is to model a population. Parametric models are often appropriate in common situations with similar mechanisms.
- Parametric models have many known and useful properties and are easy to work with. When fitting a population, only a few parameters need to be estimated: *parametric inference*.
- Sometimes one does not want to make parametric assumptions and would rather work with non-parametric models. But non-parametric models can be infinite dimensional.
- In this course, we emphasize on parametric models.

Discrete uniform X has the discrete uniform(1, N) distribution if X is equally likely to be one of $\{1, 2, \dots, N\}$.

- Sample space: $\{1, 2, \dots, N\}$
- pmf:

$$f_X(x) = \frac{1}{N}, \quad x = 1, 2, \dots, N$$

- cdf:

$$F_X(x) = \Pr(X \leq x) = \begin{cases} 0 & x < 1 \\ [x]/N & 1 \leq x < N \\ 1 & N \leq x \end{cases}$$

- moments:

$$EX = \frac{N+1}{2}$$

Bernoulli Distribution Consider an experiment where outcomes are binary (say, Success or Failure) and the probability of success is p . Define the following random variable

$$Y = \begin{cases} 1 & \text{outcome is success} \\ 0 & \text{outcome is failure} \end{cases}$$

Then, Y has a Bernoulli Distribution.

- Sample space: $\{0, 1\}$.
- pmf: $\Pr(Y = 1) = p$ and $\Pr(Y = 0) = 1 - p$. We can write this as:

$$f(y) = \Pr(Y = y) = \begin{cases} p^y(1-p)^{1-y} & y = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

- what are the cdf, mean and variance?

Binomial Distribution A $\text{Binomial}(n, p)$ random variable X is defined as the number of successes in n i.i.d. (independent, identically distributed) Bernoulli trials, each with probability p of success:

$$X = \sum_{i=1}^n Y_i, \quad Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$$

- Sample space: $\{0, 1, \dots, n\}$
- pmf:

$$f_X(s) = \begin{cases} \binom{n}{s} p^s (1-p)^{n-s} & s = 0, 1, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

- cdf:

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \sum_{s=0}^{\lfloor x \rfloor} \binom{n}{s} p^s (1-p)^{n-s} & 0 \leq x < n \\ 1 & n \leq x \end{cases}$$

- mean:

$$EX = np$$

- variance:

$$\text{Var}(X) = np(1-p)$$

Poisson Distribution The Poisson distribution was derived by the French mathematician Poisson in 1837 as a limiting version of the binomial distribution. The Poisson distribution is often used to model the number of occurrences in a given time interval. One of the basic assumptions on which the Poisson distribution is built is that, for small time intervals, the probability of an arrival is proportional to the length of waiting time. This makes it a reasonable model for situations such as waiting for a bus, waiting for customers to arrive in a bank.

The Poisson distribution has a single parameter λ , sometimes called the intensity parameter. A Poisson random variable X , takes values in the nonnegative integers with pmf

$$f_X(x) = \Pr(X = x|\lambda) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, \dots$$

To see that $\sum_{x=0}^{\infty} \Pr(X = x|\lambda) = 1$, recall the Taylor series expansion of $e^\lambda = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$. Thus

$$\sum_{x=0}^{\infty} \Pr(X = x|\lambda) = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} e^\lambda = 1$$

What is the mean and variance of X ?

$$\begin{aligned} EX &= \sum_{x=0}^{\infty} x \frac{e^{-\lambda}\lambda^x}{x!} \\ &= \sum_{x=1}^{\infty} \frac{e^{-\lambda}\lambda^x}{(x-1)!} \\ &= \lambda \sum_{x=1}^{\infty} \frac{e^{-\lambda}\lambda^{x-1}}{(x-1)!} \\ &= \lambda \sum_{y=0}^{\infty} \frac{e^{-\lambda}\lambda^y}{y!} \\ &= \lambda \end{aligned}$$

Similarly

$$\begin{aligned} EX^2 &= \sum_{x=0}^{\infty} x^2 \frac{e^{-\lambda}\lambda^x}{x!} \\ &= \sum_{x=1}^{\infty} x \frac{e^{-\lambda}\lambda^x}{(x-1)!} \\ &= \sum_{x=1}^{\infty} \frac{e^{-\lambda}\lambda^x}{(x-1)!} + \sum_{x=2}^{\infty} \frac{e^{-\lambda}\lambda^x}{(x-2)!} \\ &= \lambda + \lambda^2 \end{aligned}$$

So that

$$\text{Var}(X) = EX^2 - (EX)^2 = \lambda$$

- Sample space: $\{0, 1, \dots\}$
- pmf: $\Pr(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$
- cdf: $F_X(x) = \sum_{s=0}^x \frac{e^{-\lambda}\lambda^s}{s!}$

Hypergeometric Distribution Suppose a population of N entities is made up of two types: M of the first type and $N - M$ of the second type. Suppose we take a sample of size K . We wish to know X , the number in the sample of the first type. The probability mass function of X is given by:

$$f_X(x) = \frac{\binom{M}{x} \binom{N-M}{K-x}}{\binom{N}{K}}$$

for $x = \max(0, M - N + K), \dots, \min(M, K)$.

The sample space is defined so that all binomial coefficients are valid. We must have:

$$0 \leq x \leq K, \quad 0 \leq x \leq M, \quad 0 \leq K - x \leq N - M$$

Often $K < M$ and $K < N - M$ so the range becomes $0 \leq x \leq K$.

Hypergeometric vs Binomial We can show that the limiting form of the hypergeometric pmf is the binomial pmf

$$\begin{aligned} \Pr(s) &= \frac{\binom{M}{s} \binom{N-M}{n-s}}{\binom{N}{n}} \\ &= \frac{\frac{M!}{s!(M-s)!} \frac{(N-M)!}{(n-s)!(N-M-n+s)!}}{\frac{N!}{n!(N-n)!}} \\ &= \frac{\frac{n!}{s!(n-s)!} \frac{M!}{(M-s)!} \frac{(N-M)!}{(N-M-n+s)!}}{\frac{N!}{(N-n)!}} \end{aligned}$$

Note

$$\begin{aligned} \frac{M!}{(M-s)!} &= \frac{M(M-1)(M-2)\dots(M-s)!}{(M-s)!} \\ &= M^s \left[1\left(1 - \frac{1}{M}\right) \dots \left(1 - \frac{s-1}{M}\right) \right] \\ \frac{N!}{(N-n)!} &= N^n \left[1\left(1 - \frac{1}{N}\right) \dots \left(1 - \frac{n-1}{N}\right) \right] \\ \frac{(N-M)!}{[(N-M)-(n-s)]!} &= (N-M)^{n-s} \left[1\left(1 - \frac{1}{N-M}\right) \dots \left(1 - \frac{n-s-1}{N-M}\right) \right] \end{aligned}$$

Letting $N \rightarrow \infty, M \rightarrow \infty, \frac{M}{N} \rightarrow p$, we have

$$\begin{aligned}\Pr(s) &= \frac{\binom{M}{s} \binom{N-M}{n-s}}{\binom{N}{n}} \\ &\approx \binom{n}{s} \frac{M^s (N-M)^{n-s}}{N^n} \\ &= \binom{n}{s} \left(\frac{M}{N}\right)^s \left(1 - \frac{M}{N}\right)^{n-s} \\ &\rightarrow \binom{n}{s} p^s (1-p)^{n-s}\end{aligned}$$

In summary, we have

$$\begin{array}{lll}\text{Hypergeometric} & \rightarrow & \text{Binomial} \rightarrow \text{Poisson} \\ N \rightarrow \infty & & n \rightarrow \infty \quad \lambda = np \\ M \rightarrow \infty & & p \rightarrow 0 \\ \frac{M}{N} \rightarrow p & & np \rightarrow \lambda\end{array}$$

Geometric Distribution Consider a series of iid Bernoulli Trials with p = probability of success in each trial. Define a random variable X representing the number of trials until first success. Note X includes the trial at which the success occurs (one parameterization). Then, X has a geometric distribution.

- Sample space: $\{1, 2, \dots\}$
- pmf:

$$f(x) = \Pr(X = x) = \begin{cases} p(1-p)^{x-1} & x = 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

- cdf:

$$F(x) = \Pr(X \leq x) = \begin{cases} 0 & x < 1 \\ 1 - (1-p)^{\lfloor x \rfloor} & 1 \leq x \end{cases}$$

- Moments:

$$\begin{aligned}E(X) &= 1/p \\ \text{Var}(X) &= (1-p)/p^2\end{aligned}$$

Memoryless property. Suppose k and i are integers, and $k > i > 0$, then

$$\Pr(X > k | X > i) = \Pr(X > k - i)$$

Proof:

$$\begin{aligned}\Pr(X > k | X > i) &= \frac{\Pr(X > k)}{\Pr(X > i)} = \frac{(1-p)^k}{(1-p)^i} \\ &= (1-p)^{k-i} = \Pr(X > k - i)\end{aligned}$$

Example Suppose X is number of years you live, and X follows a geometric distribution, then

$$\begin{aligned}\Pr(\text{survive two more years}) &= \Pr(X > \text{current age} + 2 | X > \text{current age}) \\ &= \Pr(X > 2)\end{aligned}$$

This model is clearly too simple for human populations (since we do age).

Negative Binomial Distribution Still in the context of iid Bernoulli trials, define a random variable corresponding to the number of trials required to have s successes. We say $X \sim \text{Negbin}(s, p)$.

- Sample space: $\{s, (s + 1), \dots\}$
- pmf: for $x = s, s + 1, s + 2, \dots$

$$\begin{aligned}f(x) &= \binom{x-1}{s-1} p^{s-1} q^{x-s} \cdot p \\ &= \binom{x-1}{s-1} p^s q^{x-s}\end{aligned}$$

- cdf: no closed form
- Expectation: $EX = s/p$.
- Variance: $Var(X) = s(1-p)/p^2$

Notes

- Why the name? See Casella & Berger p.95.
- $X \sim \text{Negbin}(1, p)$ is the same as $X \sim \text{Geometric}(p)$
- $\text{Negbin}(n, p)$ is the same as the sum of n $\text{Geometric}(p)$ random variables

Other parameterizations The negative binomial distribution is sometimes defined in terms of the random variable Y = number of failures before the r th success. Then

- Sample space: $\{0, 1, 2, \dots\}$
- pmf

$$f(y) = \binom{r+y-1}{y} p^r q^y, \quad y = 0, 1, 2, \dots$$

- cdf: no closed form
- Expectation: $EY = r(1-p)/p$
- Variance: $Var(Y) = r(1-p)/p^2$

Negative binomial vs. Poisson The negative binomial distribution is often good for modeling count data as an alternative to the Poisson. In the previous parameterization, define

$$\lambda = \frac{r(1-p)}{p} \iff p = \frac{r}{r+\lambda}$$

Then we have

$$EX = \lambda$$

$$Var(X) = \frac{\lambda}{p} = \lambda(1 + \frac{\lambda}{r}) = \lambda + \frac{\lambda^2}{r}$$

For the Poisson we had that the variance equals the mean.

For the negative binomial, the variance is equal to the mean plus a quadratic term. Thus the negative binomial can capture overdispersion in count data.

In the previous parameterization, the pmf becomes

$$f(y) = \binom{r+y-1}{y} p^r q^y = \frac{(r+y-1)!}{y!(r-1)!} \left(\frac{r}{r+\lambda}\right)^r \left(\frac{\lambda}{r+\lambda}\right)^y$$

$$= \frac{\lambda^y r(r+1) \dots (r+y-1)}{y! (r+\lambda)^y} \left(1 + \frac{\lambda}{r}\right)^{-r}$$

Letting $r \rightarrow \infty$, we get

$$f(x) \rightarrow \frac{\lambda^x}{x!} e^{-\lambda}$$

So for large r , the negative binomial can be approximated by a Poisson with parameter $\lambda = r(1-p)/p$.

Common continuous distributions

Uniform Distribution A random variable X having a pdf

$$f(x) = \begin{cases} 1 & \text{for } 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

is said to have a *uniform distribution* over the interval $(0, 1)$.

The cdf is:

$$F(y) = \int_{-\infty}^y f(x)dx = \begin{cases} 0 & \text{for } y \leq 0 \\ y & \text{for } 0 \leq y \leq 1 \\ 1 & \text{for } y > 1 \end{cases}$$

- Unifrom; $Y \sim U[a, b]$
- sample space: $[a, b]$
- pdf:

$$f(y) = \begin{cases} \frac{1}{b-a} & \text{for } a < y \leq b \\ 0 & \text{otherwise} \end{cases}$$

- cdf:

$$F(y) = \int_{-\infty}^y f(x)dx = \begin{cases} 0 & \text{for } y \leq a \\ \frac{y-a}{b-a} & \text{for } a \leq y \leq b \\ 1 & \text{for } y > b \end{cases}$$

- moments:

$$E(Y) = (a + b)/2$$

$$Var(Y) = \frac{(b - a)^2}{12}$$

Notes

- The uniform extends to the continuous case the idea of equally likely outcomes.
- If $Y \sim U[0, 1]$, then $a + (b - a)Y \sim U[a, b]$

Exponential Distribution Denoted $X \sim Exp(\lambda)$:

- sample space: $x \geq 0$
- pdf:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- cdf:

$$F(x) = \int_{-\infty}^x f(y)dy = \begin{cases} 1 - e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

- moments:

$$E(X) = 1/\lambda$$

$$Var(X) = 1/\lambda^2$$

$$M_X(t) = \lambda/(\lambda - t), \quad t < \lambda$$

Interpretation The exponential can be derived as the waiting time between Poisson events. Suppose that the number of events in a unit interval of time follows a Poisson(λ) distribution. Then, let Y be the time until the first event.

$$\Pr(Y > t) = \Pr(0 \text{ events in } [0, t])$$

and the number of events in $[0, t]$ follows a Poisson distribution with parameter λt . Therefore,

$$\Pr(Y > t) = e^{-\lambda t}.$$

The cdf of Y is

$$F(t) = 1 - \Pr(Y > t) = 1 - e^{-\lambda t}$$

and hence the density is $f(t) = \lambda e^{-\lambda t}$.

Alternative parameterization Many books write the density as

$$f(y) = \begin{cases} \frac{1}{\theta} e^{-y/\theta} & \text{for } y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

so that $E(Y) = \theta$ and $Var(Y) = \theta^2$. In this case $\theta = 1/\lambda$ is called the *mean parameter*, while $\lambda = 1/\theta$ is called the *rate parameter*.

Memoryless property The exponential has a memoryless property, just like the geometric.

$$\Pr(Y > s + t | Y > t) = \Pr(Y > s)$$

Same interpretation as the geometric for continuous time:

- The probability of an event in a time interval depends only on the length of the interval, not the absolute time of the interval.
- The underlying Poisson process is stationary: the rate λ is constant. (In the geometric case, the probability, p of getting an event in every discrete time unit is constant).

Shifted exponential Let $X \sim Exp(\lambda)$ and $Y = X + v, v \in \mathbb{R}$. Then, Y has the *shifted exponential distribution* with pdf:

$$f(y) = \begin{cases} \lambda e^{-(y-v)\lambda} & \text{for } y \geq v \\ 0 & \text{otherwise} \end{cases}$$

Interpretation:

- $v > 0$: Event is delayed
- $v < 0$: The news of the event is delayed

Does the shifted exponential maintain the memoryless property?

Double exponential The *double exponential distribution* is formed by reflecting an exponential distribution around zero. It has pdf:

$$f(x) = \frac{1}{2} \lambda e^{-\lambda|x|}, \quad x \in \mathbb{R}$$

Laplace distribution Suppose X has the above distribution with $\lambda = 1$. Now let $Y = \sigma X + \mu, \mu \in \mathbb{R}$ (shifting) and $\sigma > 0$ (scaling). Then Y has the *Laplace distribution* with pdf:

$$f_Y(y) = \frac{1}{2\sigma} \exp\left(-\frac{|y - \mu|}{\sigma}\right)$$

with moments

$$EY = \mu, \quad Var(Y) = 2\sigma^2$$

The Laplace distribution provides an alternative to the normal for centered data with fatter tails but all finite moments.

Normal Distribution Introduced by De Moivre (1667 - 1754) in 1733 as an approximation to the binomial. Later studied by Laplace and others as part of the Central Limit Theorem. Gauss derived the normal as a suitable distribution for outcomes that could be thought of as sums of many small deviations.

- Sample space: $\mathbb{R} = (-\infty, \infty)$
- pdf: For $Y \sim N(\mu, \sigma^2)$,

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}} \quad -\infty < y < \infty$$

- cdf: There is no closed form.
- When $\mu = 0$ and $\sigma = 1$, the distribution is called *standard normal*:

$$\Phi(y) = \Pr(Y \leq y), \quad \Phi(-y) = 1 - \Phi(y)$$

- Mean:

$$EY = \mu$$

- Variance:

$$Var(Y) = E(Y - \mu)^2 = \sigma^2$$

- Higher central moments:

$$E(Y - \mu)^m = \begin{cases} \frac{m!}{2^{m/2}(m/2)!} \sigma^m & m \text{ is even} \\ 0 & m \text{ is odd} \end{cases}$$

- In particular:

$$\begin{aligned} \mu_3 &= E(Y - \mu)^3 = 0 \text{ (Skewness)} \\ \mu_4 &= E(Y - \mu)^4 = 3\sigma^4 \end{aligned}$$

- Moment generating function:

$$M_Y(t) = \exp(\mu t + \sigma^2 t^2 / 2)$$

Standardization

$$Y \sim N(\mu, \sigma^2) \iff Z = \frac{Y - \mu}{\sigma} \sim N(0, 1)$$

Shifting and scaling:

$$Z \sim N(0, 1) \iff Y = \sigma Z + \mu \sim N(\mu, \sigma^2)$$

Notes

- Normal distribution is useful in many practical settings. For example, measurement error.

- Plays an important role in *sampling distributions* in *large samples*, since the Central Limit Theorem says that the sums of independent identically distributed random variables are approximately normal
- There are many important distributions that can be derived from functions of normal random variables (e.g., χ^2 , t , F). We will briefly present the pdf's and sample spaces of these distributions.